School-Linked Social Insurance and Labor Market Effects: Evidence from Mexico*

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Abstract

In 2016, Mexico implemented a school-linked expansion of access to the national social insurance system (IMSS) for more than six million public high school and college students. In a setting where informality is widespread, understanding whether school-based access to social insurance affects early labor market attachment is empirically relevant. I estimate the impact of this policy on early labor market outcomes. The research design exploits variation in exposure across birth cohorts, interacted with pre-policy state-level informality, to identify an intent-to-treat effect that varies with initial informality. Using national labor force surveys I find that, in states with higher pre-policy informality, exposed cohorts exhibit higher labor force participation and formal employment. Quantitatively, a one-percentage-point increase in a state's initial informality is associated with a 0.8-1.3 percentage-point increase in formal employment among exposed cohorts. Effects are similar for women and men. These cross-cohort changes coincided with exposure to the policy rather than to pre-existing trends. These results document positive short-run impacts on formal attachment in markets with initially high informality.

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1 Introduction

Informal employment refers to jobs lacking legal and social protection in law or in practice. According to the International Labour Organization, more than 2 billion workers worldwide are in informal employment, representing over 60% of the workforce (ILO, 2018). While informal employment is present across all countries, it is a key characteristic of labor markets in developing countries (Ulyssea, 2020; Breza and Kaur, 2025). High rates of informality are associated with higher incidence of poverty and job insecurity (Ulyssea et al., 2018; Donovan et al., 2023; Brockmeyer et al., 2025). At the macro level, informality represents an important obstacle for tax collection and social insurance provision (Jensen, 2022; Bachas et al., 2023). Informal workers usually lack access to social protection programs that provide a safeguard against negative shocks such as disease or unemployment. Policymakers and international institutions have raised the question of whether expanding social protections outside formal employment reduces the relative value of formal contracts and thus tilts incentives toward informality. This debate has intensified with the global push toward universal social protection (USP2030), led by the World Bank Group and the International Labour Organization. The central question, therefore, is how policy design interacts with pre-existing informality to shape career trajectory formal attachment and, ultimately, the transition to formality.

In this paper, I study a school-linked social insurance policy enacted in Mexico. In 2016, students enrolled in public high schools and colleges began receiving medical benefits from the national social insurance system (IMSS) while they remained in school, independently of their employment status. Using information from the quarterly National Survey of Occupation and Employment (ENOE), I evaluate the impact of gaining access to these benefits on early career labor market outcomes of the exposed population, varying with initial informality. Early experiences in the labor market are important as there is evidence of high persistence of informality (Akay and Khamis, 2012) and of *scarring* effects of youth informality (Cruces et al., 2012) on future employment and earnings. The Mexican labor market provides a good scenario for this study, as it has a large share of informal employment (close to 60%), with significant heterogeneity across states.[1]

Using a cohort-by-state difference-in-differences design, I compare birth cohorts that were still in school in 2016 (and therefore eligible for school-linked IMSS benefits) with slightly

[1]National informality fell modestly from 57% in 2016 to 55% in 2025. Informality at the state level ranges roughly from 40% to 80%. Source: INEGI.

older adjacent cohorts serving as controls, and interact exposure with pre-policy state informality to recover an intention-to-treat differential. Because eligibility was rolled out nationwide within a short window, there is no same-age untreated counterfactual or staggered geographic adoption. The design therefore relies on cross-cohort exposure and pre-existing state conditions rather than a direct treated vs. untreated comparison. Intuitively, the design asks whether the exposure gap between adjacent cohorts is larger where informality was initially higher, net of cohort, state, and time effects.

Exposed cohorts in more informal states display higher labor force participation and formal employment. Quantitatively, each 1 percentage point increase in initial informality is associated with a 0.4-0.6 percentage points rise in participation and a 0.8-1.3 percentage points rise in formal employment among exposed cohorts. I find no detectable effects on school enrollment, overall employment, or monthly income, consistent with reallocation from informal to formal jobs rather than an expansion of total employment. Effects are similar for women and men, for students and non-students, and for individuals in households with either formal or informal household heads. Results are robust to excluding the self-employed. Evidence from targeted labor market insertion surveys of high school graduates (ENILEMS) corroborates the increase in participation and shows qualitatively similar patterns for formalization. Identification is supported by flat pre-reform cohort-informality gradients and placebo tests on cohorts too old to be treated.

The remainder of the paper is organized as follows. In Section 2, I describe the policy and the potential mechanisms. In Section 3, I briefly discuss the relevant literature. Section 4 is devoted to the description of the data, research design, and presentation of the main results. Section 5 presents results by school enrollment, gender, and an alternative definition of informality. Section 6 presents results from an alternative survey, with detailed information on the labor market outcomes of high school graduates. Section 7 concludes.

2 Expansion of Social Insurance Benefits for Students

In late 2015, Mexico established a health insurance plan for all students enrolled in public high schools and universities. Beginning in 2016, students enrolled in public schools became eligible for IMSS medical benefits while in school, independently of their employment status. Enrollment assigned each student a permanent social security number, which is required to hold a formal job in Mexico.[2] Students obtained free access to medical consultations, lab tests, medicines, certain surgical interventions, pregnancy support, and guidance on nutrition, addictions, and sexual education.

In 2016, the federal and state governments launched the "Tienes IMSS" (You Have IMSS) campaign to enroll students and promote benefit take-up. There was a widespread media campaign and big political events with the presence of the president, cabinet members, and governors of the states announcing the enrollment progress. Between March and August 2016, over six million students were registered, effectively covering the universe of students in public high schools and colleges. Although administrative efforts were coordinated by states and thus staggered, the window from launch to near-universal coverage was about five months. Given the quarterly frequency of ENOE, this rapid nationwide rollout leaves insufficient cross-state timing variation to exploit geographic staggering. The empirical strategy therefore relies on cross-cohort exposure interacted with pre-policy state conditions rather than a direct treated-versus-untreated comparison. Figure A.1 in the Appendix shows how enrollment into the program evolved. Section B in the Appendix includes additional operational details.

In the Mexican context, the core distinction between formal and informal employment is access to social insurance (principally healthcare and retirement contributions). Survey evidence indicates that health coverage is the most valued component of formal jobs.[3] Public-school enrollment rates are high: the coverage rate of the population in high school age is 74% and roughly 80% of high school students attend public institutions.[4] Thus, this policy affected a substantial portion of Mexican youth and provides a setting to study how early, school-linked access to health insurance (decoupled from holding a formal job) shapes early career labor market outcomes.

Ex-ante, this policy has an uncertain impact on the early labor market outcomes of the exposed population. I focus on three channels:

1. **Schooling margin (stay in school longer).**
   By attaching medical benefits to current enrollment in school, the policy raises the return to staying in school. If this dominates, exposed students delay labor market

[2]The social security number in Mexico is not used as the main identification number, but it is required for formal employment and remains fixed over the life cycle.

[3]*Módulo de Trayectorias Laborales (MOTRAL) 2015.*

[4]Secretaría de Educación Pública. *Principales Cifras del Sistema Educativo Nacional 2015-2016*.

entry: lower participation while enrolled, possibly higher enrollment, and ambiguous effects on formality conditional on working.

2. **Benefit substitution while enrolled (relative price of informality).**
Formal jobs typically bundle non-wage benefits B^F (IMSS coverage) with wages w^F. Informal jobs pay w^I with $B^I \approx 0$. Let b^S denote the value of school-linked benefits while enrolled. For students, the formal premium becomes:

$$(w^F - w^I) + (B^F - b^S).$$

If b^S appreciably closes the benefit gap $(B^F - b^S)$, the relative cost of informal work falls. Among currently enrolled, we could expect that participation may rise (work becomes more attractive) and the share of formal job falls (reallocation toward informal employment). Effects should dissipate after leaving school (when $b^S = 0$). Wage effects are a priori ambiguous (employers may substitute wages for benefits). This is precisely the concern raised by policymakers and international institutions when expanding social protection outside formal employment: if benefits are partially delinked from contracts, they may unintentionally subsidize informality on the margin.

3. **Salience, rights, and reduced frictions toward formality during the school-to-work transition.**
The program made IMSS benefits concrete (healthcare) and assigned a permanent social security number, lowering informational and administrative frictions to formal hiring. If students learn and value these benefits, they may seek formal jobs to retain them after school, and employers face lower onboarding frictions. We could expect higher formal employment at exit, little change in total employment, and stronger effects where pre-policy informality was high.

Given these opposing forces, the net effect is empirical.

3 Literature Review

This paper is most closely related to the literature studying the relationship between social protection programs that are not attached to employment and labor market outcomes, where there is a debate of their impact. In developed countries, the effect of welfare programs on labor supply has been extensively studied. The majority of the studies consider conditional cash transfer (CCT) programs and find evidence of a disincentive to work among beneficiaries (Moffitt, 2002; Meghir and Phillips, 2008). More related to this paper, Akosa Antwi et al. (2013) find that the Affordable Care Act's expansion of dependent coverage until age 26 resulted in fewer hours worked and a lower likelihood of working full-time jobs. For Latin America, Alzúa et al. (2013) find evidence for three CCT programs of negative but small and insignificant effects on the labor supply of the targeted population. However, they point to differentiated effects across genders and within the household. Alternatively, Gerard et al. (2021) find that an expansion of *Bolsa Familia* in Brazil increased local formal employment.

In the case of Mexico, there have been several studies that focused on the impact of the *Seguro Popular* program, which provided medical benefits to poor households. On one side, one of the main exponents of the argument that certain forms of social insurance can cause substantial increases in informality is Santiago Levy (2007; 2008). He argues that *Seguro Popular* acted as an implicit subsidy to informal employment and resulted in a significant reduction of formal jobs. Consistent with this, Bosch and Campos-Vazquez (2014) find evidence of a reduction in employers and employees formally registered. On the other hand, other authors argue against this idea, and most of the empirical literature does not find a significant impact on informality (Esquivel and Ordaz-Díaz, 2009; Arias et al., 2010; Campos-Vázquez and Knox, 2013; Azuara and Marinescu, 2013; Alonso-Ortiz and Leal, 2018). More recently, Seira et al. (2023) use administrative data and improved econometric methods, and find no evidence of a decrease in formal employment. This paper contributes to the literature by analyzing a social program that is not linked to employment but that could affect the incentives toward informality at a critical period, the start of labor market trajectories.

Additionally, this paper is related to a strand of literature that emphasizes the importance of early labor market experiences. There is widespread evidence regarding *scarring* effects of youth unemployment spells, otherwise known as "unlucky cohorts" that face adverse conditions at labor market entry (Arulampalam, 2001; Bell and Blanchflower, 2011; Schwandt and von Wachter, 2019). In the context of Latin America, Cruces et al. (2012) find that youth informal employment spells have an impact on persistent informality as well as wage penalty effects. Furthermore, Berniell et al. (2023) provide evidence of important gender differences in the region, with women from "unlucky cohorts" showing a higher employment rate and earnings in the future, consistent with the theory that women act as secondary workers in hard times. Particularly important for my study is the consideration of working while in school. Le Barbanchon et al. (2023) provide some experimental results of employment lotteries in Uruguay. They find a positive income effect two years after the program, while previous non-experimental evidence provided mixed results (Ruhm, 1997; Hotz et al., 2002; Ashworth et al., 2021).

It is also important to note that there are two different traditional schools of thought with respect to the barriers to entry to formal employment. The traditional, dualist school of thought argues in favor of segmented labor markets, based on the Harris-Todaro model (1970) and considers informal workers as being informal in an involuntary way. Alternatively, a revisionist school of thought argues in favor of integrated labor markets, where informality is caused by attempts of formal firms to reduce labor costs and increase competitiveness, and by entrepreneurs who choose to operate informally (Portes et al., 1989; Maloney, 2004; Bosch and Maloney, 2010). Therefore, in this school of thought, informal workers are informal voluntarily. Gabriel Ulyssea's work (2018; 2020) has considerably expanded our conceptual and empirical understanding of informality. He distinguishes two margins of informality: the *extensive margin*, where firms decide whether to register formally or not, and the *intensive margin*, where a formal firm can hire informal workers by keeping them "off the books".

For my conceptual framework, it is important to consider the *intensive margin* of informality. My setting foregrounds the worker's choice between formal and informal jobs at labor market entry. School-linked IMSS benefits change two features during enrollment: a benefit substitution component (some medical coverage is available regardless of formal status), which could lower the relative price of informal work while in school; and a salience/administrative frictions component (benefits become concrete and a permanent social security number is assigned), which can ease formal hiring at exit. Consistent with the relevance of the intensive margin in Mexico, formal firms post both formal and informal vacancies and adjust formalization after inspections (Samaniego de la Parra and Fernández Bujanda, 2024).

Accordingly, I test three predictions: (1) effects confined to currently enrolled youth would indicate benefit substitution; (2) effects concentrated at school-to-work transition with little change in total employment are consistent with salience/frictions; and (3) results should be stronger where pre-policy informality was higher. The evidence below aligns with (2) and (3), and is hard to reconcile with a large substitution toward informality during enrollment.

4 Impact on Labor Market Outcomes

4.1 Data

The main data source for this paper is the Mexican quarterly survey on employment called ENOE (*Encuesta Nacional de Ocupación y Empleo*).[5] This survey has information on approximately 200 thousand workers each quarter, is representative at the national and state level, and allows me to have information for both formal and informal workers. The design of the survey is a rotating panel, such that each individual is followed for 5 quarters, and each quarter, one fifth of the sample is replaced. Crucially, ENOE allows me to implement INEGI's official definition of employment informality, ensuring a consistent and comparable measurement across time and states. I adopt INEGI's employment-informality concept (worker/job status) and construct outcomes using the pre-coded survey variables.[6]

The cleanest design would randomize (or stagger exogenously) access to school-linked IMSS benefits across schools or states and track named individuals longitudinally through school exit and early careers, observing school type (public/private), eligibility, take-up and subsequent entry into the labor market. In practice, the 2016 rollout reached the universe of eligible public-school students nationwide within about five months, leaving no exploitable same-age untreated group or multi-period geographic staggering at ENOE's quarterly frequency. Moreover, ENOE's short rotating panel does not support tracking individuals from school to work at scale, and it does not identify whether a respondent attended a public vs. private high school nor student-IMSS take-up. I therefore adopt a cohort-by-state difference-in-differences approach: compare birth cohorts exposed to the 2016 program (still in school at rollout) with slightly older adjacent cohorts, and interact exposure with pre-policy state informality to recover a differential intention-to-treat effect. Intuitively, I ask whether the exposure gap between adjacent cohorts is larger in states that were initially more informal. The main analysis centers on 2019, when the first

[5]Due to the COVID-19 pandemic there is no information for the second quarter of 2020. For the period between the third quarter of 2020 and the fourth quarter of 2022 the information comes from the ENOE New Edition (ENOE^{N}). This temporary survey maintains the conceptual, statistical, and methodological design of ENOE, but combines in-person interviews with phone interviews. Throughout this period, in-person interviews represent between 79% and 100% of all interviews in ENOE^{N}.

[6]Specifically INEGI's definition of informal employment considers the following groups: self-employed and independent workers in the informal sector (economic units operating with household's resources that are not identifiable and independent of those households), self-employed workers in the agricultural sector, workers without pay, domestic workers without social insurance, employees with wages or non-wage perceptions but no social insurance, and workers with non-specified occupation position.

set of fully exposed cohorts completed high school and before the COVID-19 shock.

I complement my main results with information from a survey that measures the labor market insertion of high school graduates (aged 18-20), called ENILEMS (*Encuesta Nacional de Inserción Laboral de los Egresados de la Educación Media Superior*). This survey provides more detailed information about an individual's high school and labor market experiences. Crucially, I can identify whether they attended a public or private high school, making the identification of exposure to the policy cleaner. ENILEMS also provides information on the educational attainment of parents, whether the individuals attempted to, and were ever enrolled in a higher education institution, and whether the individuals were employed during high school. Unfortunately, this survey is not administered regularly. However, I can use the 2016 and 2019 surveys (the two most recent ones), that capture outcomes for cohorts both exposed to and unaffected by the policy. It is important to mention that ENILEMS started as a special module of ENOE and for its 2019 wave it became an independent survey but the sampling follows households identified by ENOE, making both surveys fully comparable. Given the limited number of waves and smaller sample, ENILEMS plays a corroborating role: it sharpens the exposure proxy and enables mechanism checks, but it is not the basis for identification.

4.2 Research Design

I focus on the high school component of the policy (roughly U.S. grades 10-12) for three reasons. First, the mapping from birth cohorts to school grade/age is tight: most high-school students are 15-18, so cohort of birth is an informative proxy for exposure. Second, coverage at high-school ages is high (74% of the age group enrolled, 80% in public schools), whereas college enrollment in Mexico is approximately 30% with much greater age dispersion.[7] Focusing on high school therefore reduces composition noise and improves interpretability. Third, the 2016 rollout reached universal coverage of public high school students within five months, leaving too little geographic timing variation at ENOE's quarterly frequency to credibly exploit staggered adoption. For these reasons, the birth cohort determines eligibility exposure in my design. Analyzing the college component and longer-run outcomes is left for future work.[8]

[7] Secretaría de Educación Pública. *Principales Cifras del Sistema Educativo Nacional 2015-2016*.

[8] Because the policy also applied to public universities, some individuals in the control cohorts (those just older in 2016 could have become eligible upon entering public college. ENOE does not identify public vs. private college attendance, so I cannot fully purge this "treated-controls" contamination. This attenuates cohort constrasts and biases the estimates toward zero. Consequently, my effects could be read as lower bounds on the combined high school + college schooling-linked impact.

The ideal experiment would randomly assign school-linked IMSS eligibility at the individual (or school) level and follow named individuals before and after the reform, comparing treated and otherwise identical controls of the same age, sate, and background. With such data, we could estimate a direct treated vs. untreated effect within an age cell and trace dynamics trhough school exit and early careers.

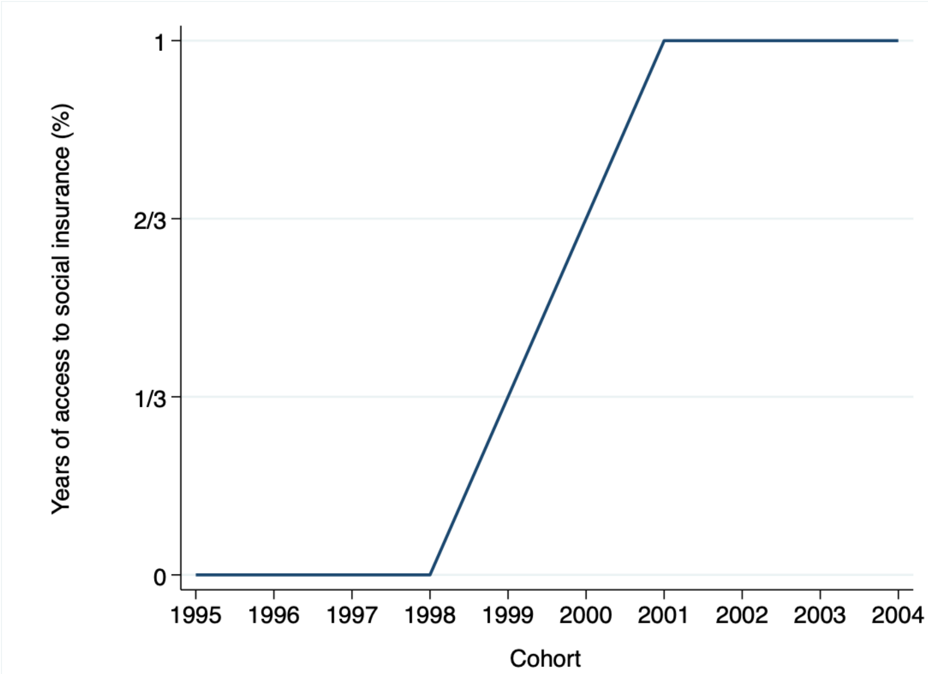
However, eligibility was expanded nationwide and almost simultaneously in 2016. Any staggered administrative adoption across states was compressed into 5 months, too short to exploit at ENOE's quarterly frequency. ENOE also does not record public vs. private high school (nor student-IMSS take-up), and its rotating panel does not follow youth long enough to observe clean transitions from school to stable employment.[9] Crucially, for the young treated cohorts we do not observe pre-policy labor outcomes because they were below labor force age before 2016. These features rule out a same-age treated vs. control comparison and a conventional before-after design.

I therefore exploit the eligibility cutoff in cohort space. Birth cohorts entering high school in 2016 (or younger) are exposed; slighlty older adjacent cohorts are unexposed cohorts. I measure exposure continuously as the share of grades 10-12 that coincided with the implementation of the policy. Figure 1 plots the exposure measure. Cohorts that graduated from high school before 2016 (i.e. were older than 18) were not exposed at all to the policy, while cohorts that were 15 or younger in 2016 entered high school after the full implementation of the program and were fully exposed. There were two cohorts that were partially exposed, corresponding to those students who only had the social insurance benefits for one or two years of their high school.

Because the reform's bite plausibly depends on local frictions and returns to formality, I interact this cohort exposure with pre-policy state informality. Figure 2 plots the pre-policy informality rate by state (2013-2015 average), showing substantial heterogeneity. I treat the pre-policy informality as an intensity shifter of the reduced-form effect, which I am unable to estimate in the conventional before-after design. Intuitively, the design asks whether the gap between adjacent cohorts is larger in states that were initially more informal, after absorbing cohort, state, and time effects, as well as common shocks.

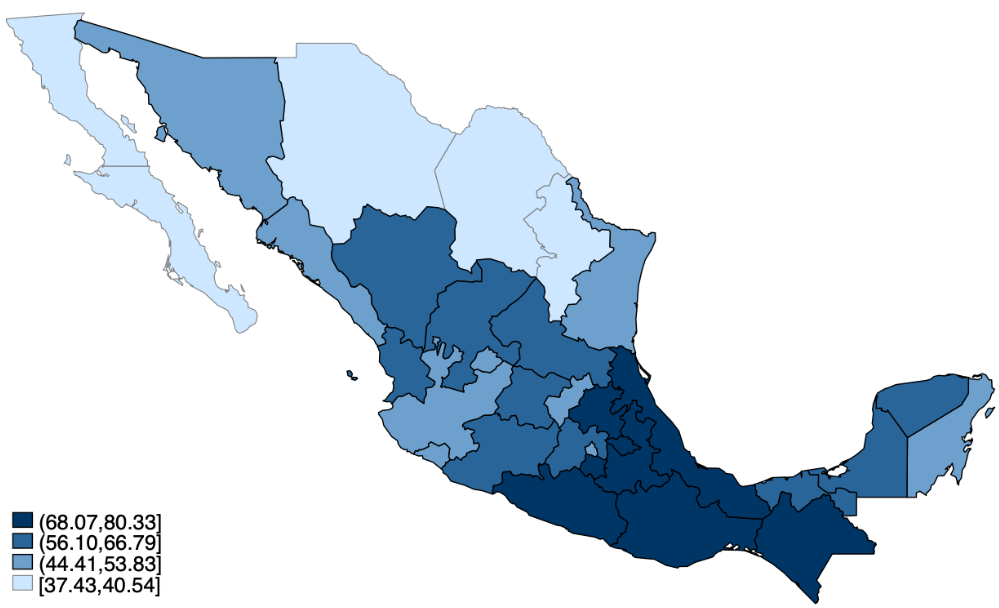
[9] Even if school type were observed, public-private comparisons would not necessarily be credible counterfactuals: arguably they could differ systematically in selection, resources, and trajectories.

Figure 1: Exposure to high-school-linked social insurance by cohort



Note: the figure shows the exposure to the policy by cohort, defined as the percentage of years of high school when students had access to social insurance benefits.
Source: Author's calculation.

Figure 2: Pre-policy informality rate by state (%)
2013-2015 average



Source: Author's calculations with data from ENOE.

This strategy identifies an intention-to-treat (ITT) differential. That is, how the policy's effect scales with initial informality, rather than an aggregate average effect or a within-age treated vs. untreated effect. This mirrors the logic in Duflo (2001) and Bleakley (2010): exploit timing by cohort and variation in pre-policy intensity, and demonstrate that only the exposed cohorts exhibit breaks aligned with exposure.

Outcomes are constructed at the cohort-state-quarter level. Intuitively, the difference-in-differences (DiD) design is outlined in Table 1. For simplicity, consider two state groups by pre-policy informality: low (L) and high (H). Within each group compare exposed cohorts (still in high school at rollout) to unexposed adjacent cohorts. Let Y_g^E and Y_g^U denote mean outcomes for group $g \in \{L, H\}$. Each $\Delta_g = Y_g^E - Y_g^U$ is the exposed-unexposed gap within a given state environment. The difference-in-differences estimand across state groups is $\beta = \Delta_H - \Delta_L$. This is the ITT differential: how the exposure effect varies when moving from low to high pre-policy informality. In the continuous specification used in the paper "L vs. H" becomes the slope of Δ_s with respect to $I_{s,pre}$ (pre-policy informality at the state level), and exposure is measured continuously (share of grades 10-12 under the policy.

Table 1: Cohort-by-state DiD with intensity

	Low pre-policy informality	High pre-policy informality
Exposed cohorts	Y_L^E	Y_H^E
Unexposed adjacent cohorts	Y_L^U	Y_H^U
Within-state cohort difference	$\Delta_L = Y_L^E - Y_L^U$	$\Delta_H = Y_H^E - Y_H^U$
Estimand (ITT differential): $\beta = \Delta_H - \Delta_L$		

The unexposed controls are older than exposed cohorts, so any comparison $Y^E - Y^U$ mixes (i) a life-cycle/cohort difference and (ii) the policy effect on the younger, exposed cohort. The design removes the life-cycle component by differencing across state environments. Let D_s denote the baseline cohort/age gap (unrelated to the policy) and τ_s denote the policy effect on exposed cohorts within state s. We can express the within-state cohort difference as:

$$\Delta_s = D_s + \tau_s.$$

We only observe the sum. However, taking the difference between high- and low-pre-policy informality environments:

$$\beta = \Delta_H - \Delta_L = (D_H - D_L) + (\tau_H - \tau_L)\,.$$

If, absent the policy, the baseline cohort/age gap D_s is the same across state environments (or, more generally, uncorrelated with pre-policy informality after conditioning on fixed effects and controls, then $D_H - D_L = 0$ and the DiD estimand identifies $\tau_H - \tau_L$: the differential policy effect. Equivalently, we allow exposed cohorts to be younger than unexposed, we only require that age/cohort profile of outcomes does not systematically differ by pre-policy informality once we net out cohort,state, and time effects and standard controls. That is the parallel trends in cohort space assumption necessary for identification of the ITT differential. In other words, we can interpret the identifying assumption underlying my analysis as assuming that in the absence of the policy, the average change in the unexposed cohorts among low informal states represents the counterfactual change in the exposed cohorts in high informal states across labor market outcomes.

There are several features of my empirical design that make this asssumption plausible. First, I use slightly older adjacent cohorts as controls, minimizing life-cycle spacing. Second, cohort fixed effects absorb national age/cohort profiles; state fixed effects absorb time-invariant differences; time fixed effects absorb national shocks. I consider alternative specifications that include state-level trends to consider differentiated shocks. Third, I estimate β_k (slope of outcome vs. pre-policy informality for each birth cohort) and overlay exposure timing. Identification requires β_k flat for unexposed cohorts and breaking only when exposure is positive. Fourth, I consider placebo cohorts. For cohorts too old to be exposed, $\Delta_H - \Delta_L$ should be zero. Results are robust across gender, school attendance, formality of the household head, and focusing on workers excluding the self-employed.

4.3 Main Results

4.3.1 Long Differences

First, I simplify the analysis and aggregate the different cohorts into two groups: those who were fully exposed to the policy and those who were not exposed at all. I compare the main labor market outcomes (participation rate, employment rate, formality rate, monthly income) across these groups of cohorts for each state to obtain long difference estimates. Therefore, the basic equations to be estimated is:

$$Y_{s,E} - Y_{s,U} = \beta I_{s,pre} + X_{s,pre}\Gamma + \alpha + \varepsilon_s,$$

where Y is a labor market outcome of state s, and the time subscripts E/U indicate whether the cohorts were fully exposed or not to the policy. The preceding informality rate is $I_{s,pre}$, defined as the average state informality rate between 2013 and 2015. The X variables are a set of controls measuring conditions at the state level before the implementation of the policy, and α is a constant term. The main outcomes of interest are the participation, employment, and formality rates, and monthly labor income. For the set of controls I include state GDP per capita in 2013, mean educational attainment in 2015 and regional controls, grouping the 32 states into 8 different regions, outlined in figure A.4 of the Appendix. The equation is estimated using weighted least squares with data from the four quarters of the employment survey of 2019, where the weights are defined by the square root of cell sizes to take into account the different precision of the estimations of the cohort means. The coefficient β is the ITT differential in long differences: the extra exposed–unexposed gap associated with a 1-p.p. increase in initial informality.

The long differencing estimates are presented in Table 2. Panel A shows the estimates with the preceding informality rate as the only independent variable, and Panel B includes the full set of controls. From the mean differences between cohorts fully exposed and not exposed at all, we can see that the exposed cohorts have significant lower participation, and formality rates, as well as average income. These patterns are expected as fully exposed cohorts are younger. Table A.1 in the Appendix includes additional descriptive statistics.

From the estimated coefficients in panel B, we can interpret that one additional percentage point in the preceding informality rate is associated with an increase of 0.59 percentage points in the participation rate, an increase of 0.15 percentage points in the employment rate, and an increase of 0.87 percentage points in the formality rate among the exposed cohorts. The estimated impact on labor income, while positive, is not statistically significant. The sample of income results is smaller due to higher non-response rates in this question of the survey. These reduced form results suggest that in states with high informality rates, cohorts that were exposed to the policy experienced an increased participation in the labor market, with higher employment and increased formality, relative to cohorts in less informal states.

Table 2: Cross Cohort Long Differences in 2019

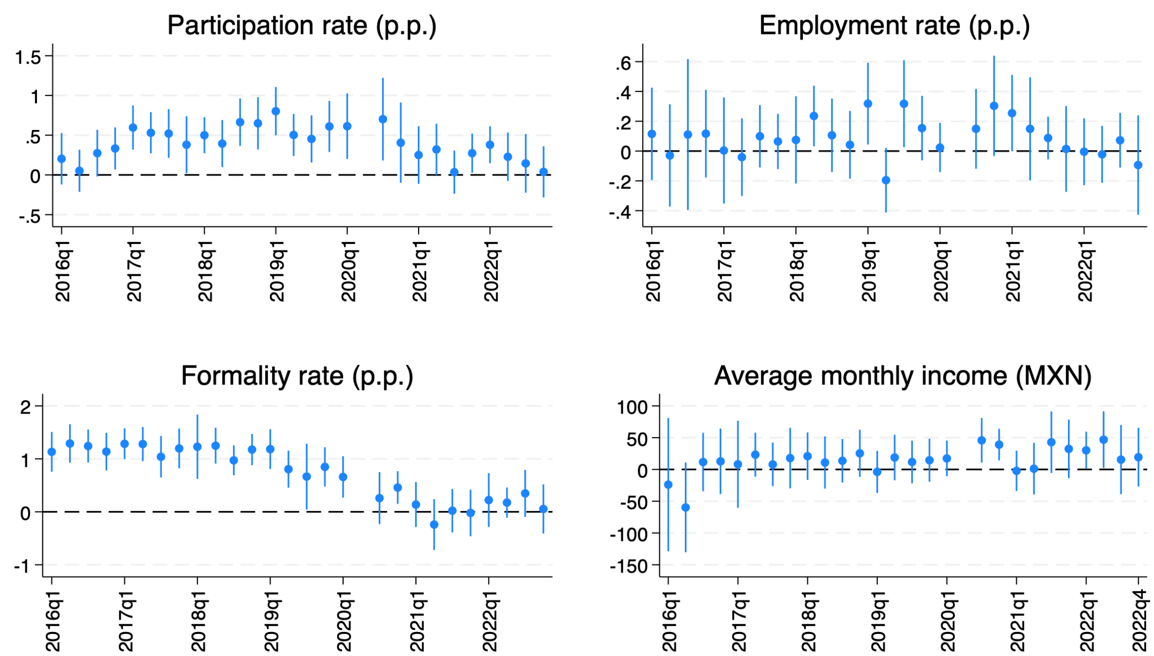
	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Panel A				
Pre-policy informality rate	0.50***	0.16***	0.79***	30.07***
	(0.03)	(0.02)	(0.04)	(2.75)
Panel B				
Pre-policy informality rate	0.59***	0.15*	0.87***	10.32
	(0.07)	(0.07)	(0.10)	(7.87)
Log GDP per capita in 2013	6.04***	1.58	1.91	-122.39
	(1.14)	(1.47)	(1.41)	(110.52)
Educational attainment in 2015	-3.29***	-0.37	2.25	-350.76**
	(0.90)	(1.02)	(1.38)	(112.90)
Regional controls	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
Mean dependent variable	-37.89	0.08	-32.75	-1,930.18
Observations	128	128	128	128

Bootstrap standard errors in parentheses. The square root of cell sizes is used to construct weights for the observations. Cell sizes are identical across specifications, except for the income one as we restrict to individuals reporting strictly positive income.
***p<0.01, **p<0.05, *p<0.1

The analysis is done with information from 2019 to evaluate the medium-term effects of the policy for the fully exposed cohorts in 2016, right before the onset of the COVID-19 pandemic in 2020. To consider the full picture, Figure 3 plots the estimated coefficients ($\hat{\beta}_t$) of the long differences equation for each quarter t from 2016 to 2022. We can observe that the positive effects on the participation and formality rate are relatively consistent from 2016 to 2019. After the COVID-19 pandemic, the estimated coefficients for both of these indicators are close to zero and no longer statistically significant.

It is not clear how to interpret these estimates as the pandemic severely disrupted labor markets. There is evidence that in Latin America there was an initial drop in participation, employment, and informality followed by a rebound in informality, with the informal sector acting as an important margin of adjustment (Alvarez and Pizzinelli (2021), Leyva and Urrutia (2023)). Furthermore, Osuna-Gómez (2023) finds that people that entered the labor market after the Great Recession lost formal employment during the pandemic at higher rates than previous cohorts. It could be the case that the pandemic shock reverted any effects that were driven by the school-linked social insurance policy on labor market outcomes. However, there are additional concerns for more recent estimates as we are tracking cohorts at the state level and not individuals, and we could worry about potential migration patterns and readjustments between different local labor markets. The employment and income estimates are mostly small and not significant.

Figure 3: Estimated coefficients of pre-policy informality rate on cross cohort long differences, $\hat{\beta}_t$



Note: No available data for 2020q2.
Source: Author's calculations with data from ENOE.

4.3.2 Panel Estimates

The long difference estimates transparently summarize medium-run gaps between fully exposed and unexposed adjacent cohorts across states with different initial informality. Their limitation is that they do not, by construction, demonstrate that such gaps were not already drifting with pre-policy conditions. To address this, I turn to a panel framework that (a) uses all quarters from 2016 to 2019 to absorb common shocks with survey-wave fixed effects and (b) follows Bleakley's cohort-space logic: for each birth cohort I recover the cross-state gradient with respect to pre-policy informality and show that this gradient is flat for cohorts too old to be exposed and turns on only at the eligibility cutoff. In addition, I report robustness with state-specific linear time trends to soak up differential growth by state unrelated to the policy.

First, to incorporate cohorts that were partially exposed to the policy I estimate pooled regressions of the form:

$$Y_{kst} = \tilde{\beta} I_s \times \mathrm{Exp}_k + \delta_k + \delta_s + \delta_t + \sum_i \left(x_s^i \times \mathrm{Exp}_k \right) \gamma_i + \nu_{skt},$$

where Exp_k is exposure to the policy for cohort k as defined by figure 1, I_s is the pre-policy informality rate in state s, the x_s^i are state-specific controls, and $\delta_k, \delta_s, \delta_t$ are fixed effects for cohort, state, and quarter, respectively.

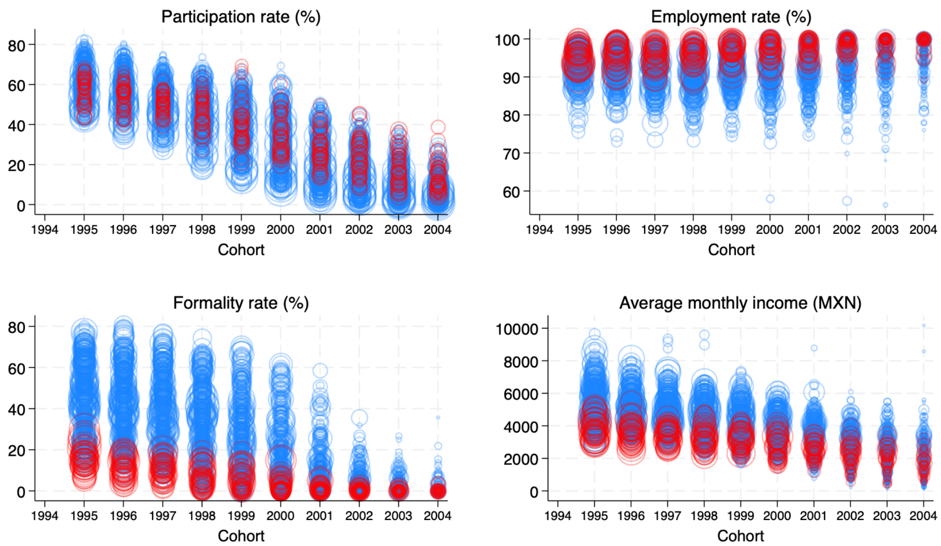
Second, I compare changes in outcomes by cohort across states to assess the contribution of the policy to the observed changes. The regressions to be estimated are:

$$Y_{kst} = \beta_k I_{s,pre} + \delta_k + X_s \Gamma_k + \nu_{skt},$$

where β_k is a cohort specific coefficient on pre-policy informality, X_s is a vector of state controls, and δ_k and Γ_k are cohort-specific intercept and slope coefficients. By estimating the equation through weighted least squares for each cohort, I generate a series of estimates across cohorts. The series of cohort estimates helps to show graphical evidence that the results are capturing the impact of the policy by comparing it with the pattern of the exposure measure.

Before presenting the estimates of the pooled and cohort-specific regressions, figure 4 plots the summary statistics for the panel constructed by cohort in states and quarters between 2016 and 2019. Each observation represents the value of the dependent variable for each cohort-state-quarter tuple. The size of the observations is determined by the size of the population represented. I single out in red the three poorest states (Guerrero, Oaxaca, and Chiapas), which also happen to be the southernmost states. We can observe expected patterns in the data, such as lower participation rates and lower formality rates for younger cohorts. Among the poorest states it is interesting to observe that for the oldest cohorts analyzed, the participation rate is lower relative to richer states but for the youngest cohorts, relative to richer states, the participation rate is higher. It is also worth noting that the employment rate is consistently higher for the poorest states across all cohorts, but the quality of jobs within these states is worse, as seen by the lower formality rate and lower monthly income. Table A.1 in the Appendix shows additional descriptive statistics of the sample.

Figure 4: Summary statistics of panel by cohort across states and quarters, 2016 - 2019



Note: Each observation represents the value of the dependent variable for each cohort by state-quarter. The red observations denote the three poorest states (Guerrero, Oaxaca, and Chiapas). The size of the observations is determined by the size of the population for each cohort-state-quarter tuple.
Source: Author's calculations with data from ENOE.

Table 3 presents the estimates for the pooled regressions. The regressions are estimated through weighted least squares with the weights constructed from the square root of cell size. Standard errors are clustered at the state level. We can observe that these estimates are consistent with the results of the long differencing. The magnitude of the estimate for the participation rate decreased but remains significant. The estimate for the employment rate is very similar, whereas the estimate for the formality rate increased to almost 1.3 percentage points associated with an increase of 1 percentage point in the initial informality rate. This means that taking into account the panel structure and the partial exposed cohorts increases the differential positive impact of the policy on the formalization of exposed cohorts in high informal states relative to low informal states. Income effects are statistically detectable but economically small (≈ 1.4 USD per month), consistent with a reallocation toward formal jobs without large short-run gains in earnings.

A natural concern is that labor-demand fluctuations (national or state-specific) could differentially affect youth and mimic treatment heterogeneity. The panel specification mitigates this along several layers, and I add targeted robustness checks. Time fixed effects absorb nationwide demand movements (e.g. macro cycles, fuel-price spike), ensuring identification does not come from aggregate time shocks. State fixed effects soak up permanent gaps in demand, institutions, and composition across states (industrial structure, enforcement). Cohort fixed effects net out nationwide age/cohort profiles in youth demand or school-to-work transitions that are common across states. The identifying variation compares exposed vs. unexposed adjacent cohorts within the same state and quarter, scaled by the state's pre-policy informality rate. Any state-quarter demand shock that affects young cohorts symmetrically cancels in this within-state contrast. Additionally, I augment the panel with state-specific linear time trends so that smooth state-level growth is absorbed. Table A.2 in the Appendix shows that the results are materially unchanged. The stability of the estimated coefficients rules out concerns for a demand-driven explanation.

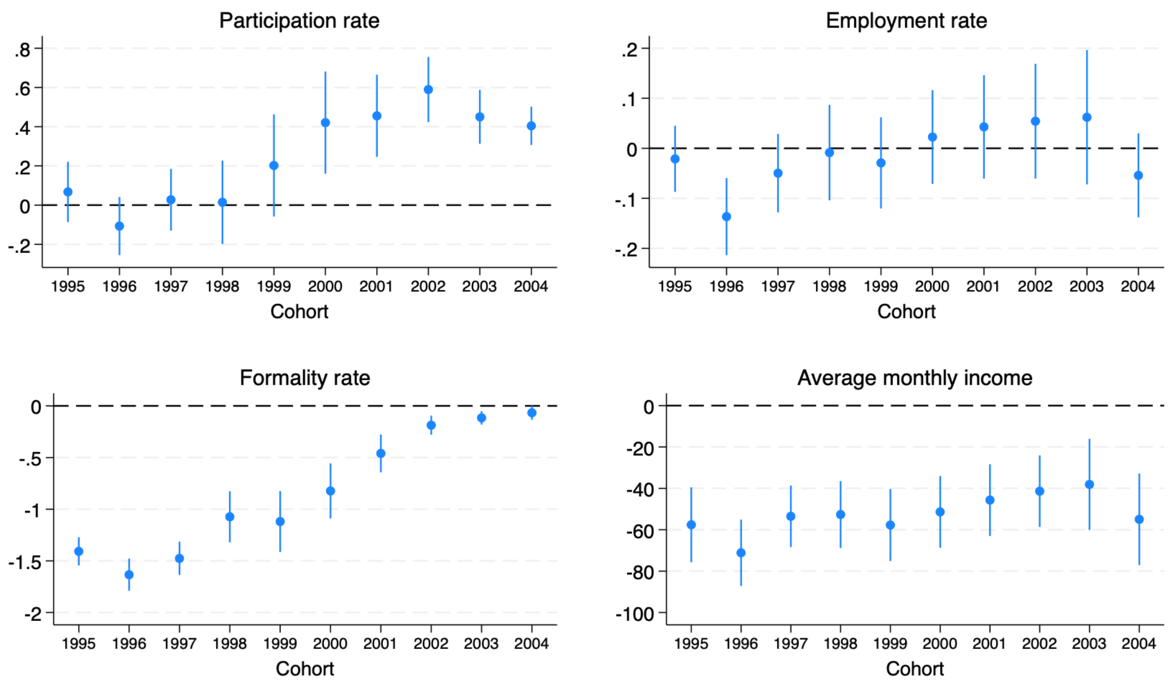
Table 3: Panel Estimates of the Effect of Exposure to Policy, 2016 - 2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Pre-policy informality	0.44***	0.15***	1.28***	21.33***
rate × Exposure	(0.05)	(0.03)	(0.06)	(7.05)
Log GDP per	2.01	0.43	3.67**	-61.73
capita in 2013 × Exposure	(1.01)	(1.02)	(1.01)	(114.12)
Educational attainment	-0.33	1.86**	1.86	-72.11
in 2015 × Exposure	(0.67)	(0.55)	(1.22)	(86.32)
Cohort FE	✓	✓	✓	✓
State FE	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
Mean dependent variable	33.31	93.84	17.99	4,299.21
Observations	5,094	5,091	5,091	5,046

Standard errors clustered by state in parentheses. The square root of cell sizes are used as weights for the observations. Cell sizes are identical across specifications, except for the income one as I restrict to individuals reporting strictly positive income.
***p<0.01, **p<0.05, *p<0.1

Figure 5 plots the cohort-specific estimates on the pre-policy informality rate for the different outcomes of interest. The pattern of estimates is broadly consistent with the exposure to the policy, as defined in figure 1. The patterns are more clear for the participation and formality rates. This graphical evidence points to the relevance of the policy explaining the results found in the long-differencing and pooled regression analyses.

Figure 5: Cohort-specific estimates on pre-policy informality rate, $\hat{\beta}_k$



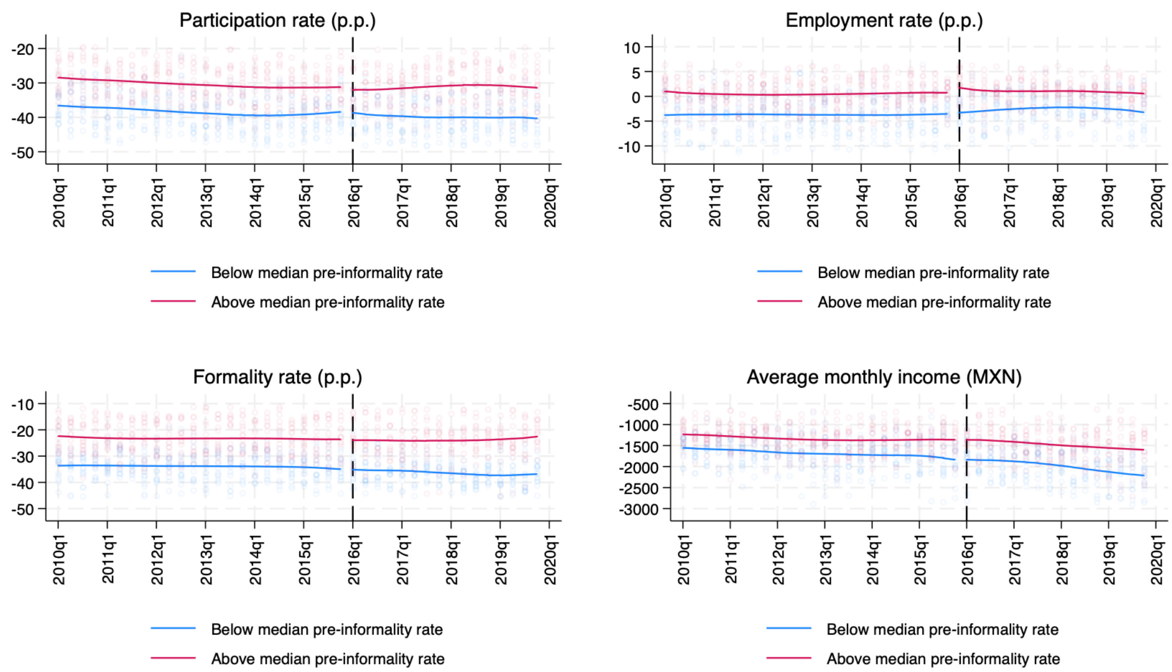
Source: Author's calculations with data from ENOE.

4.4 Tests to Identifying Assumption

In this section I analyze the validity of the identifying assumption underlying previous results. The main threat to identification, which could provide an alternative explanation for the main results, would be convergence. It could be the case that even in the absence of the policy, we would have observed an increase in the participation and formality of younger cohorts in high informal states as these very informal, poor areas catch up with the more formal, richer states.

As a first approach to test the identifying assumption, figure 6 plots the trends in the long differences between the correspondent treatment (15-18) and control (21-24) age groups of the main analysis since 2010. For each quarter, I calculate the long differences across the main outcomes for each state. I then divide states between those below and above median pre-policy informality rate, and plot the trends of the long differences before and after the implementation of the policy as estimated through a locally weighted scatterplot smoothing (LOWESS). A visual analysis of the pre-trends does not identify significant differences, suggesting that the differences across cohorts in low informal states could be an appropriate counterfactual for the differences across cohorts in high informal states. Additionally, Table A.3 in the Appendix shows that the long differences results are robust to controlling for the long differences three years prior.

Figure 6: Long Differences Trends (difference between 15-18 and 21–24 cohorts by state)



Source: Author's calculations with data from ENOE.

As a falsification test aimed at convergence, I replicate the panel design on cohorts too old to be affected. I assign placebo exposure to the 1995-1998 cohorts and use the 1991-1994 cohorts as controls. If differential catch-up in high=informality states were driving the main estimates, the coefficient on the interaction between pre-policy informality and exposure should be positive and of comparable magnitude to the baseline (younger cohorts in high-informality states "catching up" to older cohorts in low-informality states). Table 4 shows the opposite: placebo coefficients are near zero and/or negative. For participation and formality the estimates are statistically indistinguishable from zero or point to divergence among these older, untreated cohorts, and the confidence intervals exclude effects as large as the baseline. This pattern undercuts the convergence hypothesis and supports a causal differential ITT interpretation.

Table 4: Placebo Regressions

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Panel A: Long Differences, 2019				
Pre-policy informality rate	-0.11	-0.01	-0.24**	-7.16
	(0.06)	(0.04)	(0.09)	(7.93)
Full set of controls	✓	✓	✓	✓
Mean dependent variable	-12.47	-2.17	-8.07	-1,050.38
Observations	128	128	128	128
Panel B: Pooled Regressions, 2016 - 2019				
Pre-policy informality rate × Exposure	-0.09**	0.04*	-0.10	10.83**
	(0.03)	(0.02)	(0.05)	(3.23)
Full set of controls	✓	✓	✓	✓
Mean dependent variable	60.42	93.00	40.45	5,288.04
Observations	4,096	4,096	4,096	4,096
Panel C: Pooled Regressions, 2016				
Pre-policy informality rate × Exposure	-0.01	0.08**	0.14	10.11**
	(0.04)	(0.03)	(0.09)	(2.94)
Full set of controls	✓	✓	✓	✓
Mean dependent variable	54.39	91.88	34.81	4,490.38
Observations	1,024	1,024	1,024	1,024

Standard errors in parentheses (bootstrap standard errors for long differences and clustered standard errors by state for pooled regressions). The square root of cell sizes is used to construct weights for the observations. Cell sizes are identical across specifications, except for the income one as we restrict to individuals reporting strictly positive income.
***p<0.01, **p<0.05, *p<0.1

5 School Enrollment and Robustness

An important consideration is whether this policy had an impact on the schooling decisions of individuals. The new medical benefits associated with being enrolled in a public high school or college can increase school enrollment and attainment to the detriment of participation in the labor market at younger ages, which could be a desirable outcome from both the perspective of the individual and society as a whole. The results of the previous section of an increased participation rate among exposed cohorts in high informal states suggest that this is not the case. In this section, I explore more thoroughly the impact on school enrollment as well as differentiate the main results by school enrollment to better understand the mechanisms at play. I then look at the heterogeneous results and robustness to excluding the self-employed.

5.1 School enrollment

First, I explore the impact of the social insurance expansion policy on school enrollment rates of the cohorts analyzed. Table 5 shows estimates of the long differences and panel equations presented in sections 4.3.1 and 4.3.2, using school enrollment rates as the dependent variable. There is no significant evidence of an impact of the policy on schooling rates when comparing exposed and unexposed cohorts in high-informal states relative to low-informal states.

Table 5: Long Differences and Panel Estimates of School enrollment Rates

	(1) Long Differences	(2) Panel Estimates
Pre-policy informality rate	-0.21*	-0.12
	(0.09)	(0.09)
Log GDP per capita in 2013	-9.36***	-1.76
	(2.01)	(1.60)
Educational attainment in 2015	4.41**	-2.55
	(1.38)	(1.38)
Regional controls	✓	
Cohort FE		✓
State FE		✓
Quarter FE	✓	✓
Mean dependent variable	46.42	63.97
Observations	128	5,120

Standard errors in parentheses (bootstrap standard errors for long differences and clustered standard errors by state for pooled regressions). The square root of cell sizes are used as weights for the observations. For column (3) the independent variables are interacted with the policy exposure function as in Table 2.
***p<0.01, **p<0.05, *p<0.1

Now, I distinguish the main results of the previous section between workers going to school at the time of the survey and those not going. Table 6 presents the estimates of long differences on the main outcomes of the labor market distinguishing by school enrollment. Panel A shows the main results, considering the full sample, which are exactly those presented previously in Table 2. Panel B shows the long differences estimates using the sample of people not going to school and Panel C shows the results using the sample of students at the time of the survey. Approximately 53% of the full sample studied is enrolled in school. We can see that the results for participation rate, formality rate, and monthly income are higher among the sample that is going to school. In the case of the formality rate, one percentage point increase in the pre-policy informality rate is associated with an increase of 0.95 percentage points in the formality rate of the cohorts affected by the policy.

To see this graphically, figure 7 shows the estimated coefficients of the pre-policy informality rate on the main outcome variables distinguishing by school enrollment. This figure presents the results from both the long differences and panel specifications. For the case of the panel specification the coefficient corresponds to the interaction of the pre-policy informality rate and the exposure function. We confirm that the estimated coefficients are higher for the subsample of people that are in school, across the specifications and main outcomes (with the exception of the employment rate).

We can conclude from this that the main results are consistent across the subsamples of people that are enrolled and not enrolled in school at the time of the survey. In particular, these results provide suggestive evidence that students affected by the social insurance expansion policy are not more likely to be employed informally, despite the reduction in the relative cost of an informal job while also being enrolled in school. This is more relevant as there is evidence that there was an increase in the participation rate among exposed cohorts in states with higher informality.

It is uncertain whether it is desirable to have more individuals start working while they study. Employment spells while studying could provide valuable training experiences. On the other hand, it could deter from further investing in schooling (Atkin, 2016). However, if anything, it is desirable to have students work formally vs. informally at the beginning of their labor market trajectories as there are negative employment and earning effects associated with youth informality spells.

Table 6: Cross Cohort Long Differences by School Enrollment, 2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Panel A: Full Sample				
Pre-policy informality rate	0.59***	0.15*	0.87***	10.32
	(0.07)	(0.07)	(0.10)	(7.87)
Log GDP per capita in 2013	6.04***	1.58	1.91	-122.39
	(1.14)	(1.47)	(1.41)	(110.52)
Educational attainment in 2015	-3.29***	-0.37	2.25	-350.76**
	(0.90)	(1.02)	(1.38)	(112.90)
Regional controls	✓	✓	✓	✓
Mean dependent variable	-37.89	0.08	-32.75	-1,930.18
Panel B: Sample Not In School				
Pre-policy informality rate	0.37***	0.10	0.83***	19.31
	(0.11)	(0.09)	(0.13)	(10.44)
Log GDP per capita in 2013	-1.02	2.43	2.32	-242.00
	(2.52)	(1.69)	(1.76)	(164.33)
Educational attainment in 2015	-2.65	-1.80	2.59	-119.53
	(1.59)	(1.26)	(1.58)	(158.17)
Regional controls	✓	✓	✓	✓
Mean dependent variable	-17.25	-1.13	-31.43	-1,493.67
Panel C: Sample In School				
Pre-policy informality rate	0.56***	0.20	0.95***	29.13
	(0.09)	(0.11)	(0.14)	(17.08)
Log GDP per capita in 2013	4.06*	1.20	-2.91	-359.45
	(1.59)	(1.89)	(2.91)	(385.59)
Educational attainment in 2015	-0.79	1.52	5.07*	24.63
	(1.40)	(1.66)	(2.57)	(292.01)
Regional controls	✓	✓	✓	✓
Mean dependent variable	-21.95	0.68	-30.78	-2,120.30
Observations	128	128	128	128

Bootstrap standard errors in parentheses. The square root of cell sizes are used as weights for the observations.
***p<0.01, **p<0.05, *p<0.1

Figure 7: Estimates on pre-policy informality rate by school enrollment

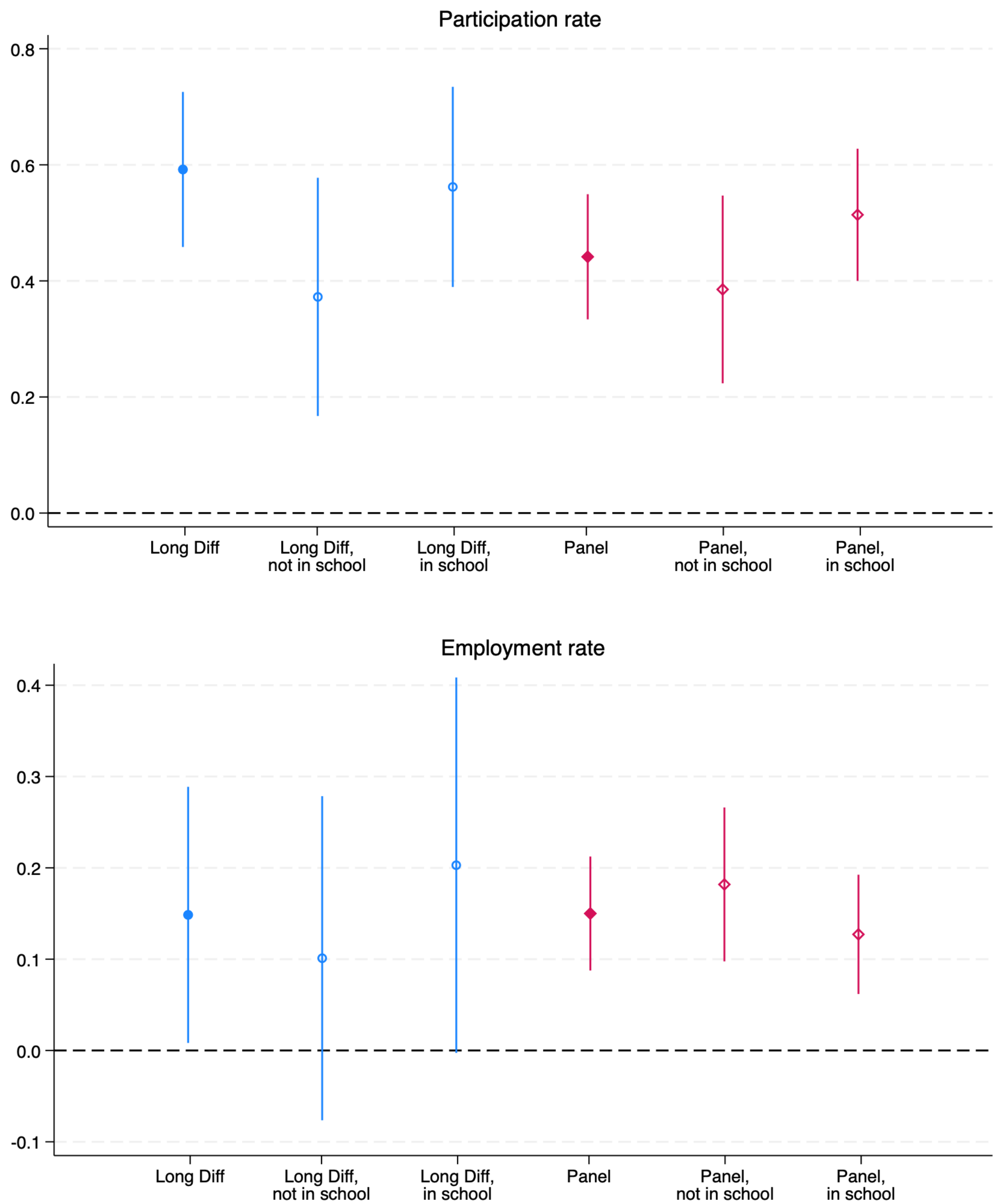
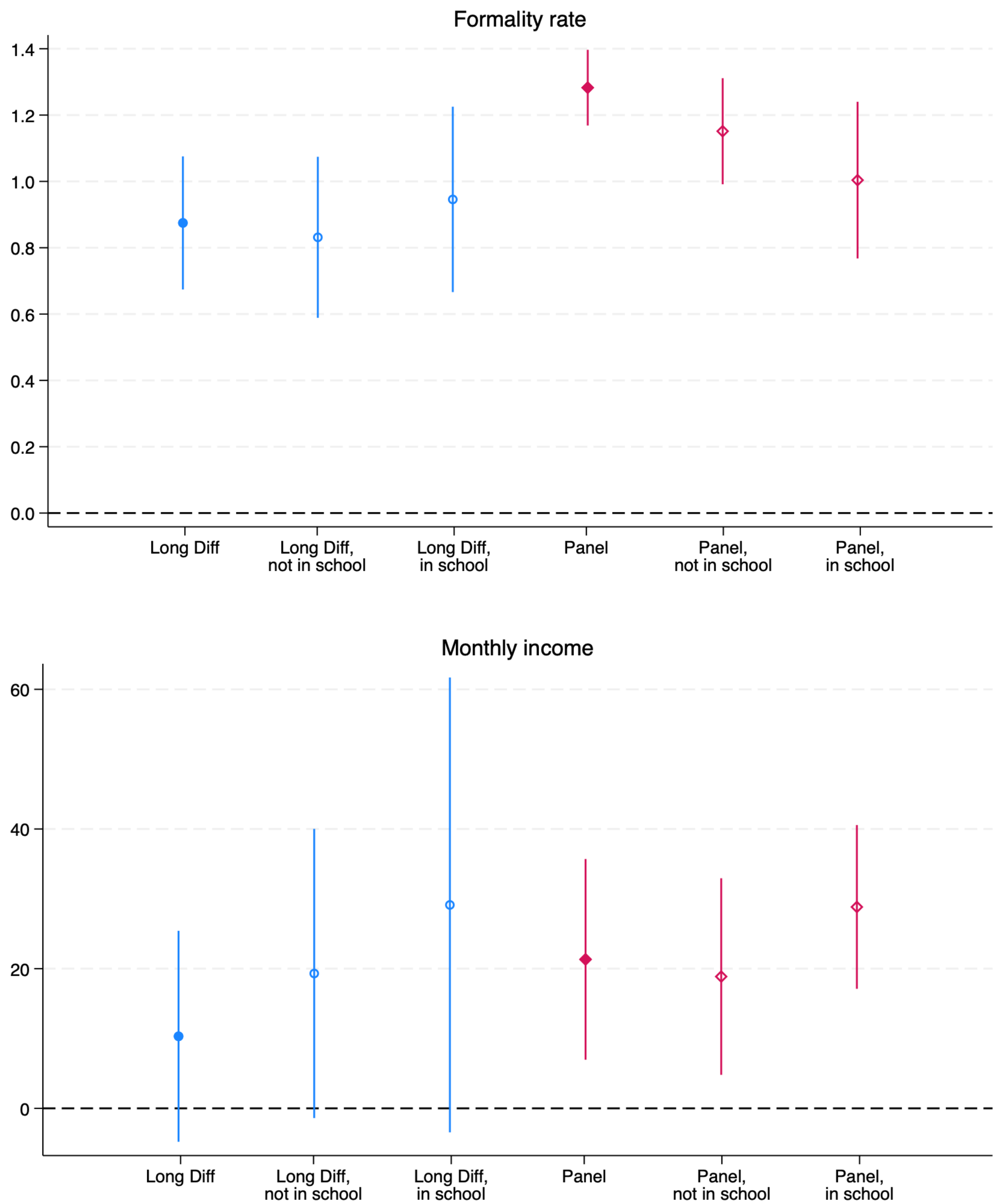


Figure 7: Estimates on pre-policy informality rate by school enrollment (cont.)

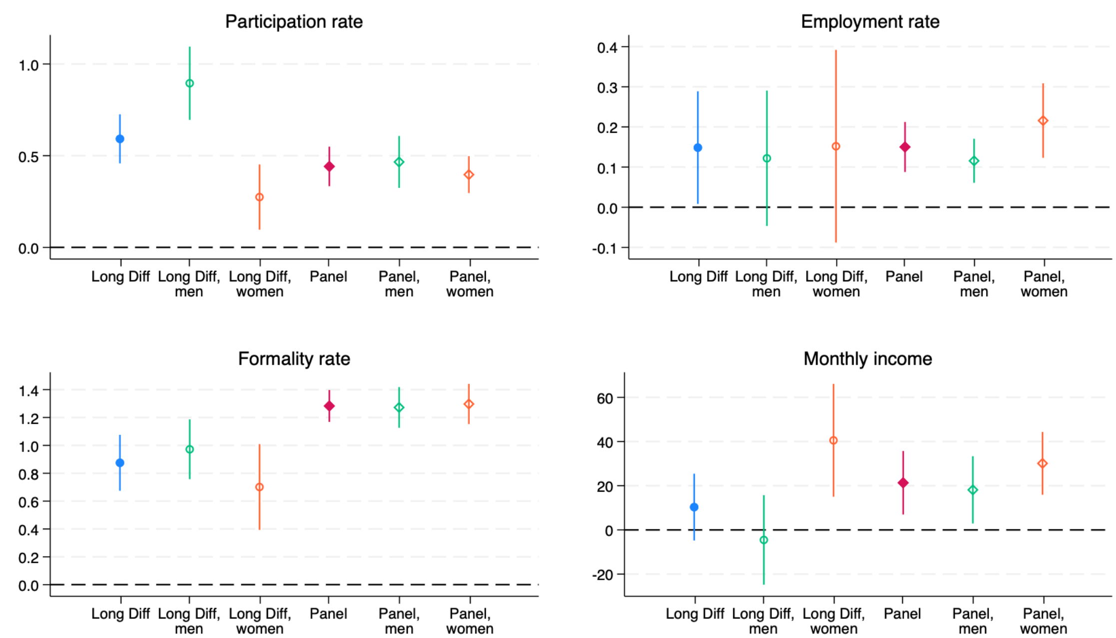


Source: Author's calculations with data from ENOE.

5.2 Heterogeneous Results

I distinguish the main results among the sample of men and women to explore any differential impact of the policy by gender. The sample is roughly evenly split between men and women. Figure 8 presents the graphical results of the coefficient on the pre-policy informality rate between specifications and by gender.

Figure 8: Estimates on pre-policy informality rate by gender

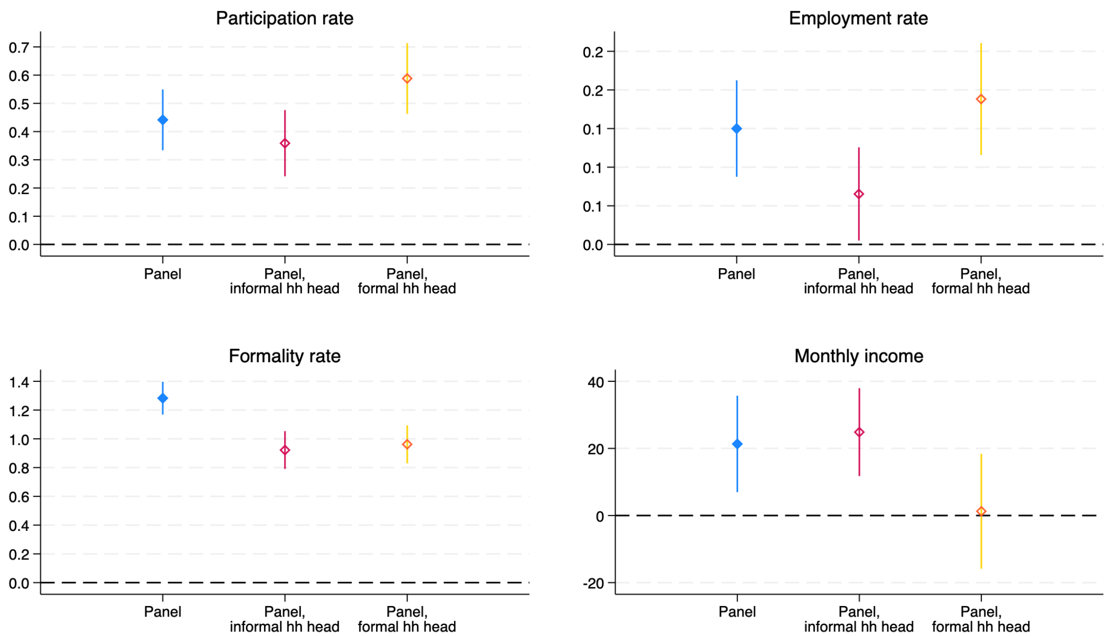


Source: Author's calculations with data from ENOE.

From these results, we cannot observe a stark difference in the impact of the policy between men and women. However, it is the case that the point estimates for the participation and formality rate are higher for men than for women, whereas the point estimates for the employment rate and monthly income are higher for women. Overall, I do not find significant differences in the impact of the policy by gender.

I further make a distinction by the formality of the head of household. Figure 9 shows that while the impact on the participation rate is larger for individuals with a formal head of household, the formalization effect is similar regardless. The fact that the results are not local to fully informal households highlights the importance of having access to social insurance benefits directly, not only through a parent.

Figure 9: Estimates on pre-policy informality rate by formality of head of household (hh)



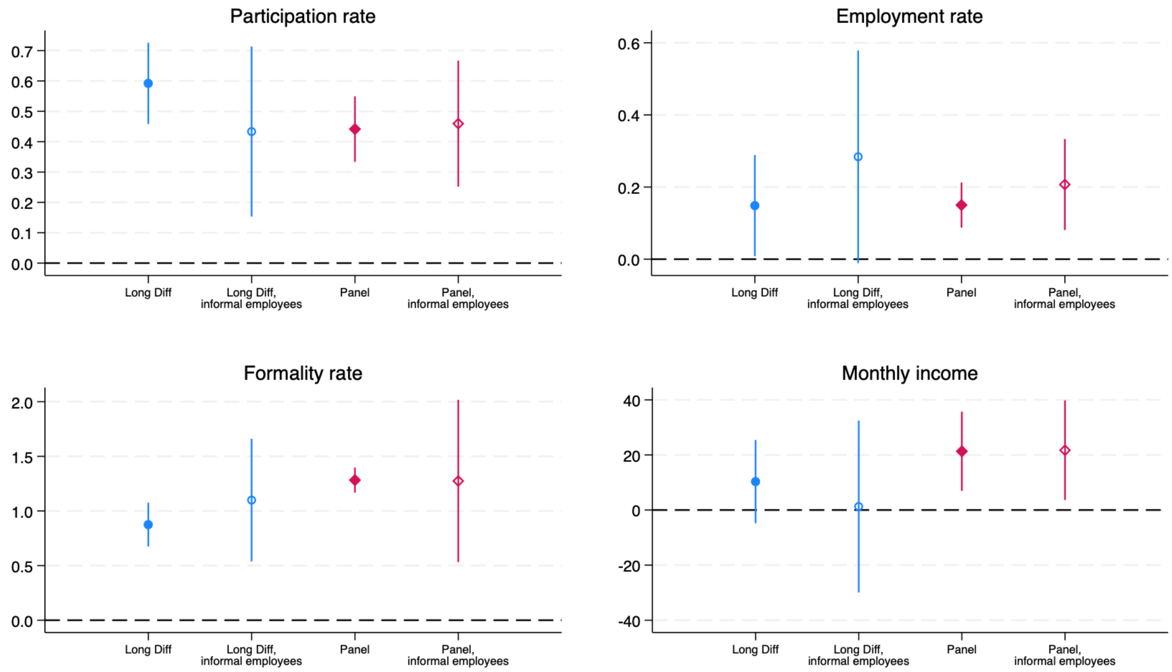
Source: Author's calculations with data from ENOE.

5.3 Alternative Sample of Informal Workers

Up to this point, I have been using the official definition of informal employment established by the Mexican National Institute of Statistics and Geography (INEGI). This definition includes both employees and self-employed individuals. I explore how the main results change if we focus only on subordinate employment, i.e., employees without access to social insurance. I use this alternative sample that only considers employees for the pre-policy period as well as for the main period of analysis.

Figure 10 presents the main results of the long differences and the panel specifications with full controls along with the results using the alternative sample of workers, considering only employees. As expected, the alternative estimates on the pre-policy informality rate become noisier but are similar in magnitude to the main estimates. The panel estimates suggest a positive and significant impact of the policy on the participation and formality rates among exposed cohorts in states with high levels of informality.

Figure 10: Estimates on pre-policy informality rate, alternative definitions of informality



Source: Author's calculations with data from ENOE.

6 Labor Market Insertion

I complement my main results with information from ENILEMS, a specialized survey of 18-20 year old high school graduates. This dataset offers richer detail on students' high school histories and subsequent labor market experiences. Crucially, it distinguishes between public and private high schools, facilitating a cleaner identification of individuals' policy exposure, and it is fully comparable with ENOE. I use the 2016 and 2019 waves of the survey, which allows me to compare cohorts exposed to, and unaffected by the policy. ENILEMS is conducted in the third quarter of the year, after the school year has finalized. Therefore, the cohorts surveyed in 2016 were not exposed to the policy (or minimally during their last year of high school), and the cohorts surveyed in 2019 were partially and fully exposed to the policy. Specifically, the two waves of ENILEMS contain information for cohorts born between 1996 and 2001.

Table 7: Labor Market Insertion Data (ENILEMS) - Summary Statistics

	(1) 2016 survey	(2) 2019 survey	(3) Difference
Female	0.54	0.53	-0.00 (-0.14)
Public high school	0.83	0.83	0.01 (0.71)
Had job during high school	0.16	0.20	0.04*** (4.61)
Continues in first job	0.35	0.41	0.06*** (3.96)
Participation rate	0.42	0.56	0.13*** (10.32)
Employment rate	0.89	0.92	0.03*** (2.76)
Formality rate	0.34	0.32	-0.02 (-1.09)
Real monthly income	3,066	3,765	699*** (6.25)
Attempted higher education	0.40	0.45	0.05*** (4.18)
Ever enrolled in higher education	0.26	0.34	0.08*** (6.77)
Either parent with higher education	0.21	0.23	0.02 (1.45)
Observations	7,191	7,799	

t-statistic in parentheses. Survey weights used.
***p<0.01, **p<0.05, *p<0.1

Table 7 presents summary statistics for both waves of the survey. Each wave contains information for more than 7 thousand individuals, representative of a population of between 2.1 and 2.9 million. Around 83% of high school graduates were enrolled in a public school across both waves. There were some significant changes in the labor market outcomes of high school graduates between both surveys. There was a significant increase in the share of individuals that worked during high school and those that continued working in their first job. We observe a sizable increase in the participation rate (from 42% to 56%). Additionally, there were also increases in the employment rate and the real monthly income. Along schooling outcomes there were significant changes as well, with an increase of the share of high school graduates that attempted and were ever enrolled in a higher education institution.

I estimate panel specifications similar to those presented in section 4.3.2 and Table 3. I take advantage of the information of type of high school (public vs. private) to refine my exposure policy function. I only considered as exposed those cohorts that were born after 1998 and that went to a public high school. To account for labor market and family background conditions across cohorts, states, and waves; I control for state GDP, the share with at least one parent with higher education, and the share that worked during high school.

Table 8: Labor Market Insertion Panel Estimates, 2016 - 2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income	(5) Ever enrolled in higher ed
Pre-policy informality	1.31*	0.15	1.27	103.52**	-1.17
rate $\times$ Exposure	(0.57)	(0.28)	(0.70)	(33.84)	(1.04)
Log state GDP	-0.10*	-0.02	-0.08	-6.25**	0.10
$\times$ Exposure	(0.04)	(0.02)	(0.05)	(1.97)	(0.08)
Share with higher ed	-0.53	0.46	0.02	-53.73	-0.55
parents $\times$ Exposure	(0.40)	(0.31)	(0.37)	(30.24)	(0.72)
Share that worked	0.12	-0.02	-1.06***	-31.39	1.38**
in HS $\times$ Exposure	(0.25)	(0.14)	(0.27)	(22.76)	(0.47)
Public/Private HS FE	✓	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓
Survey wave FE	✓	✓	✓	✓	✓
Mean dependent variable	50.17	90.70	33.89	4,212.60	31.15
Observations	376	373	319	372	374

Standard errors clustered by state in parentheses. The square root of cell sizes are used as weights for the observations. Cell sizes are identical across specifications, except for the income one as we restrict to individuals reporting strictly positive income.
***p<0.01, **p<0.05, *p<0.1

Table 8 presents the main results of the panel estimates using the labor market insertion data from ENILEMS. The interaction of the refined exposure policy and the pre-policy informality rate is associated with a significant increase in the participation rate, at the 10% significance level. The estimates for the employment and informality rates are noisily estimated and not significant, but very similar to the main results presented in Table 2. Cohorts that were exposed to the policy in less formal states and had a higher share of individuals working during high school show a decrease in formality rate, compared to unexposed cohorts in more formal states. This result provides suggestive evidence to the existence of the mechanism that considers the new acquired medical benefits through schooling as a complement of employment compensations. However, exposed cohorts in less formal states with a higher share of individuals working during high school also show a higher share of enrollment in higher education institutions. This result goes in line with the first potential mechanism discussed, where the policy increased the value of (public) education.

7 Conclusion

Informal employment remains a defining feature of labor markets in developing countries and is especially prevalent at labor market entry. Because early informal spells can scar future formal attachment and earnings, understanding whether school-linked access to social insurance alters early choices is policy-relevant.

This paper exploits a cohort-by-state design to study Mexico's 2016 policy that made public high-school students eligible for IMSS medical benefits while enrolled. Comparing exposed and slightly older adjacent cohorts and allowing effects to vary with pre-policy state informality, I estimate a differential intention-to-treat: the policy's impact is larger in initially more informal states. Exposed cohorts in high-informality states exhibit higher participation and markedly higher formal employment. Point estimates imply that a 1-pp increase in initial informality is associated with a 0.8–1.3-pp rise in formal employment among exposed cohorts. Scaled by a one-standard-deviation shift in initial informality, effects are economically large (10.46-16.99-pp increase in formality among exposed cohorts). I find no effects on school enrollment and only very small changes in income, consistent with reallocation from informal to formal jobs rather than an expansion in total employment. Effects are similar for women and men and for students and non-students, and robust to excluding the self-employed.

Taken together, the evidence is most consistent with a salience/administrative-frictions channel operating at the school-to-work transition (benefits become concrete; a permanent social-security number reduces hiring frictions), and inconsistent with the view that delinked benefits subsidize informality during enrollment. The results should be read as lower bounds on the schooling-linked impact, since some "control" cohorts may later gain eligibility in public college. They also pertain to the early-career margin and a pre-pandemic window (2016–2019).

Policywise, these findings suggest that well-designed, school-linked health coverage can complement formal labor-market entry without inducing a shift toward informality, particularly where informality is initially high. A full welfare evaluation would integrate labor market effects with health benefits and longer-run outcomes; assessing the college component and post-2019 dynamics are fruitful next steps. More broadly, advancing formalization will require understanding both workers' early-career decisions and firms' hiring margins in high-informality environments. Finally, given the relative stability of informality rates in countries such as Mexico, it is relevant to assess which policies foster formalization and which hinder it. Improving the working conditions of millions of people in the developing world remains one of the main challenges from a policy perspective.

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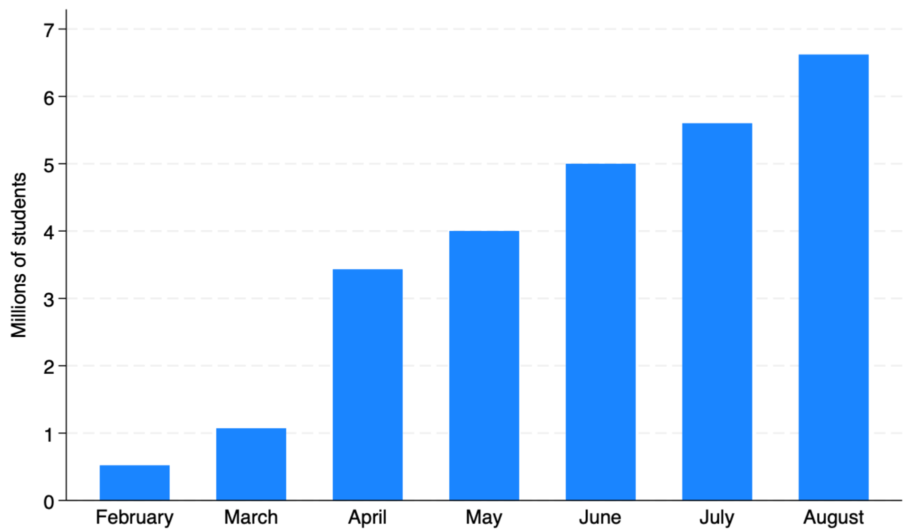
A Additional Material

Figure A.1: Evolution of cumulative students' enrollment to social insurance, 2016



Source: IMSS. *Informe de Labores y Programa de Actividades 2016 - 2017*.

Figure A.2: "Tienes IMSS" campaign flyer

Source: IMSS. Official Twitter account (April 18, 2016).

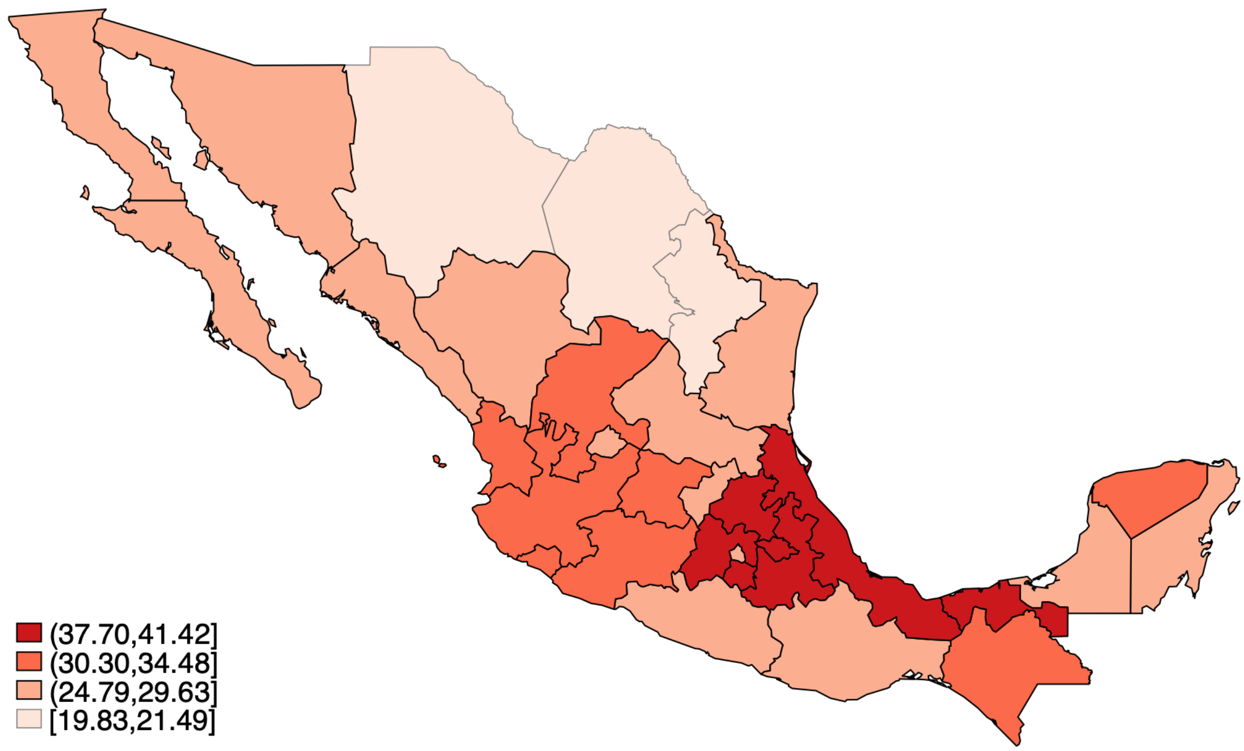
Figure A.3: Political event announcing the affiliation of students to IMSS in Mexico City

Source: IMSS. *El Seguro Médico para estudiantes es un escudo para prevenir enfermedades crónicas: Mikel Arriola.*

Figure A.4: Regions of Mexico

Source: *Regiones de México.* Northwest region includes Baja California, Baja California Sur, Chihuaha, Durango, Sinaloa and Sonora. Northeast region includes Coahuila, Nuevo León and Tamaulipas. West region includes Colima, Jalisco, Michoacán and Nayarit. East region includes Hidalgo, Puebla, Tlaxcala and Veracruz. Central north region includes Aguascalientes, Guanajuato, Querétaro, San Luís Potosí and Zacatecas. Central south region includes Mexico City, Mexico State and Morelos. South west region includes Chiapas, Guerrero, and Oaxaca. Southeast region includes Campeche, Quintana Roo, Tabasco and Yucatán.

Figure A.5: Alternative definition of pre-policy informality rate by state (%), only employees
2013-2015 average



Source: Author's calculation using data from ENOE.

Table A.1: Descriptive Statistics

	Long Differences (2019)		Panel Estimates (2016-2019)	
	(1) Exposed cohorts	(2) Unexposed cohorts	(3) Exposed cohorts	(4) Unexposed cohorts
Participation rate	0.23	0.61	0.21	0.53
Employment rate	0.93	0.93	0.95	0.92
Formality rate	0.08	0.41	0.07	0.35
Monthly income	3,501	5,384	2,906	4,602
School enrollment	0.77	0.31	0.79	0.40
Pre-policy informality rate	0.59	0.58	0.59	0.58
Log GDP per capita in 2013	11.68	11.72	11.67	11.70
Educational attainment in 2015	9.10	9.18	9.10	9.16
Age	16.05	22.02	15.60	20.48
Men	0.50	0.50	0.49	0.50
Formal household head	0.48	0.50	0.47	0.49
Northwest region	0.13	0.13	0.13	0.14
Northeast region	0.10	0.10	0.09	0.10
West region	0.12	0.12	0.12	0.12
East region	0.15	0.15	0.15	0.15
Central north region	0.12	0.12	0.12	0.12
Central south region	0.21	0.22	0.21	0.22
Southwest region	0.11	0.09	0.11	0.10
Southeast region	0.06	0.06	0.06	0.06
Observations	128	128	5,155	3,456

Mean values for each variable are reported, weighted by survey weights. Columns (1) and (2) correspond to the long differences estimates reported in Table 2. Columns (3) and (4) correspond to the panel estimates reported in Table 3. For the long differences estimates, exposed cohorts only include fully exposed cohorts; for the panel estimates exposed cohorts include partially exposed cohorts as defined by the exposure measure plotted in figure 1.

Table A.2: Panel Estimates Robustness, 2016 - 2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Pre-policy informality	0.45***	0.15***	1.29***	24.52***
rate × Exposure	(0.06)	(0.03)	(0.06)	(8.29)
Log GDP per	2.05	0.44	3.73**	-64.72
capita in 2013 × Exposure	(1.05)	(1.08)	(1.05)	(131.27)
Educational attainment	-0.32	1.85**	1.92	-58.32
in 2015 × Exposure	(0.71)	(0.58)	(1.29)	(101.38)
Cohort FE	✓	✓	✓	✓
State FE	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
State-linear time trend	✓	✓	✓	✓
Mean dependent variable	33.31	93.84	17.99	4,299.21
Observations	5,094	5,091	5,091	5,046

Standard errors clustered by state in parentheses. The square root of cell sizes are used as weights for the observations. Cell sizes are identical across specifications, except for the income one as I restrict to individuals reporting strictly positive income.
***p<0.01, **p<0.05, *p<0.1

Table A.3: Long Differences Robustness, 2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Pre-policy informality	0.58***	0.08	0.75***	7.69
rate	(0.09)	(0.09)	(0.17)	(14.11)
Log GDP per	5.95***	0.97	0.78	-145.37
capita in 2013	(1.39)	(1.50)	(2.01)	(146.30)
Educational attainment	-3.27***	-0.23	2.51	-345.61**
in 2015	(0.97)	(1.01)	(1.48)	(119.06)
Regional controls	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
Long Diff in 2015	0.02	0.15	0.28	5.62
	(0.16)	(0.12)	(0.31)	(21.13)
Mean dependent variable	-37.89	0.08	-32.75	-1,930.18
Observations	128	128	128	128

Bootstrap standard errors in parentheses. The square root of cell sizes is used to construct weights for the observations. Cell sizes are identical across specifications, except for the income one as we restrict to individuals reporting strictly positive income.
***p<0.01, **p<0.05, *p<0.1

B Operational Details of the School-Linked Social Insurance Expansion

- **Enrollment channel (2016 cohort): assisted mass enrollment.** In early 2016, federal and state authorities coordinated with public high schools to batch-enroll students. Schools transmitted rosters identified by CURP (the national population registry code), which IMSS used to assign or validate the student's permanent social security number (NSS). This step was largely automatic from the student's perspective. No individual application was required to be assigned/validated an NSS in 2016 for those on school rosters.

- **Access to services (activation/clinic choice): minor residual steps.** To actually use healthcare services, students typically needed to confirm or choose a primary clinic (UMF) and, in some cases, complete a brief registration/verification. These are low-stakes, one-off steps (often handled by schools or the student) and do not reverse eligibility.

- **Cost reduction relative to status quo ante.** Before 2016, obtaining an NSS as a youth without a formal job involved an application and document gathering; the 2016 campaign eliminated that transaction cost at scale by pre-assigning/validating the NSS off administrative records (via CURP and school rosters).

- **Post-2019 practice (for context only)**. Later cohorts generally request their NSS individually (no longer batch-assigned), which is also why my empirical window centers on 2016–2019.